

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
 Section:

1. **Angles and velocities in collisions.** (60 points) Consider again the pion decay $\pi^+ \rightarrow \mu^+ + \nu$ from lecture. The pion has a mass of $m_\pi = 140$ MeV. Suppose the initial-state pion has an energy of 1 GeV in the lab frame.

- (a) If the outgoing muon is created along the pion's direction of motion, what is the velocity (in natural units) of the outgoing muon in the lab frame? What is the energy of the neutrino in the lab frame, and what is the direction of its momentum with respect to the pion in the lab frame?

We can use the results from discussion in the pion rest frame and then boost to the lab frame. In the pion rest frame the muon has 4-vector $p = (E_\mu, \sqrt{E_\mu^2 - m_\mu^2})$ where $E_\mu = \frac{m_\pi^2 + m_\mu^2}{2m_\pi}$ (this is the result from discussion 11 problem 1.1 with $m_2 = 0$). Plugging in numbers, $p = (110.13 \text{ MeV}, 29.87 \text{ MeV})$. The boost factor to the lab frame is $\gamma = E_\pi/m_\pi = 1 \text{ GeV}/(140 \text{ MeV}) = 7.14$, and thus $\beta = \sqrt{\gamma^2 - 1}/\gamma = 0.99$. Multiplying by the appropriate Lorentz matrix, we find the muon 4-vector in the lab frame $p = (998 \text{ MeV}, 992 \text{ MeV})$. Using the suggestion from lecture to find the velocity, $v = p/E = 992/998 = 0.994$. In the pion's rest frame, the neutrino is emitted antiparallel to the muon and has energy $E_\nu = \frac{m_\pi^2 - m_\mu^2}{2m_\pi}$ (from lecture 11), so its 4-vector is $k = (E_\nu, -E_\nu)$. Boosting this gives $k' = (2.1 \text{ MeV}, -2.1 \text{ MeV})$, so the energy of the neutrino is 2.1 MeV, the same as we would have found by energy conservation in the lab frame (up to rounding). The neutrino is still going backwards in the lab frame, because as a massless particle, we can never boost to its rest frame, so if it is moving in the $-x$ direction in one frame, it is still moving in the $-x$ direction after any boost along the x -axis.

- (b) Now suppose the muon has a lab frame energy of 900 MeV. What are the angles of the neutrino and muon with respect to the pion in the lab frame? Draw a diagram of the decay, labeling the angles and giving their numerical values in degrees.

Let the pion have 4-momentum $P = (E_\pi, p_\pi, 0, 0)$. Without loss of generality, we can consider the neutrino and muon to be in the xy -plane, so the muon 4-vector is $p = (E_\mu, p_\mu \cos \theta, p_\mu \sin \theta, 0)$ where $p_\mu = \sqrt{E_\mu^2 - m_\mu^2}$, and the neutrino is $k = (E_\pi - E_\mu, (E_\pi - E_\mu) \cos \phi, (E_\pi - E_\mu) \sin \phi, 0)$ (note that we know the neutrino energy by energy conservation). 4-momentum conservation gives $P = p + k$. Rearranging to $P - k = p$ and squaring both sides, we have

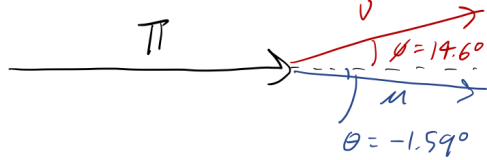
$$P^2 + k^2 - 2P \cdot k = m_\mu^2 \implies m_\pi^2 - 2(E_\pi - E_\mu)(E_\pi - p_\pi \cos \phi) = m_\mu^2.$$

Since $p_\pi = \sqrt{E_\pi^2 - m_\pi^2} = 990 \text{ MeV}$ and $E_\pi - E_\mu = 100 \text{ MeV}$, we can plug in numbers and solve for $\cos \phi$ to find $\cos \phi = 0.968$, so $\phi = 14.6^\circ$. Equating y -components of momentum gives

$$p_\mu \sin \theta + (E_\pi - E_\mu) \sin \phi = 0 \implies \theta = -1.59^\circ.$$

We can check that the x -component of momentum is also conserved:

$$p_\mu \cos \theta + (E_\pi - E_\mu) \cos \phi = (893.7 \text{ MeV})(0.9996) + (100 \text{ MeV})(0.9677) = 990 \text{ MeV} = p_\pi.$$



2. **Photon collisions.** (40 points) Morin exercise 3.36 (p. 157). *Hint: when equal angles are involved, you can often make good progress using conservation of individual components of momentum, before applying any 4-vector tricks.*

Since the initial photon has no y -component of momentum, the resulting y -components of momentum must be equal and opposite. Likewise, since the angles are equal, the x -components of momentum for the two outgoing particles must be equal. The initial x -momentum was just E , so both the mass and the outgoing photon have $p_x = E/2$. Therefore the magnitudes of their momenta are also the same, equal to $|p| = \frac{E}{2 \cos \theta}$. Since $|p|$ is also the energy of the outgoing photon, conservation of energy gives the final energy of mass m as $E_m = E + m - |p|$. Using $E_m^2 = |p|^2 + m^2$, we have

$$\begin{aligned} (E + m - |p|)^2 &= |p|^2 + m^2 \\ (E + m)^2 + |p|^2 - 2(E + m)|p| &= |p|^2 + m^2 \\ E^2 + 2Em - 2(E + m)|p| &= 0 \\ |p| &= \frac{E^2 + 2Em}{2(E + m)} \end{aligned}$$

Plugging in $|p| = \frac{E}{2 \cos \theta}$, we have finally

$$\cos \theta = \frac{E + m}{E + 2m}.$$

When $E \ll m$, $\cos \theta \rightarrow m/(2m) = 1/2$, so $\theta \rightarrow 60^\circ$.